Chapter 2

Counting and Modular Arithmetic

2.1 Basic Counting Principles

Two elementary principles act as —building blocks— for all counting problems. The first principle says that the whole is the sum of its parts; it is at once immediate and elementary.

2.1.1 Additive Principle (Sum Rule)

If a task can be done in one of two or more mutually exclusive ways, then the total number of ways to do the task is the sum of the number of ways to do each way.
Formula: If m is the number of ways to perform task A, and n is the number of ways to perform task B (A and B cannot happen at the same time), then

$$\text{Total ways} = m + n$$

Example 2.1. A student can choose 1 course from either 3 math courses or 4 science courses:

$$\text{Total options} = 3 + 4 = 7$$

The Addition Principle

For a collection of m disjoint sets with n_1 elements in the first, n_2 elements in the second, ..., and n_m elements in the m-th, the number of ways to choose one element from the collection is

$$n_1 + n_2 + \cdots + n_m$$

Example 2.2. Fast Machine Repair, Inc., offers two categories of service contracts. The first is a full-service contract that provides parts and services as needed for a fixed annual fee. The second is a service-on-demand contract that charges a small annual fee and then charges separately for services by the hour and for parts as they are needed. To satisfy different kinds of customers, Fast Machine Repair offers three levels of full-service contracts and two levels for service-on-demand service. These two sets of choices are disjoint. How many different contracts should be available so that no matter what option the customer may choose, the appropriate contract is ready for signing?

Solution: Since there are three choices for the full-service option and two-choices for the hourly charge option, and because these two sets of choices are disjoint, total number of choices is

$$\begin{aligned}(\#\text{Service options}) &= (\#\text{Full-service options}) + (\#\text{Hourly charge options}) \\ &= 5\end{aligned}$$

Example 2.3. How many ways can you choose a 9, a red card with value greater than 9, or a black card with a value less than 6 from a standard deck of cards? (In this example, an Ace is a high card.)

Solution. The choices can be put into three disjoint sets. The Addition Principle says that there are

$$4 + 10 + 8 = 22$$

possible choices.

2.1.2 Multiplicative Principle (Rule of Product)

If a task is made up of sequential steps, and each step can be performed independently of the others, then the total number of ways to perform the task is the product of the number of ways to perform each step.
If a task consists of a sequence of independent steps, and each step can be performed in a certain number of ways, then the total number of ways to complete the task is the **product** of the number of ways for each step.
Formula: If a task involves two steps, with m ways to do the first and n ways to do the second, then

$$\text{Total ways} = m \times n$$

Example: A password consists of 2 letters followed by 3 digits. Each letter has 26 options, and each digit has 10 options:

$$\text{Total passwords} = 26 \times 26 \times 10 \times 10 \times 10 = 676{,}000$$

The Multiplication Principle

Let $m \in \mathbb{N}$. For a procedure of m successive distinct and independent steps with n_1 outcomes possible for the first step, n_2 outcomes possible for the second step, ..., and n_m outcomes possible for the mth step, the total number of possible outcomes is

$$n_1 \cdot n_2 \cdots n_m$$

Example 2.4. A university offers six sections of a physics course. Each student registered for the physics course also registers for 1 of 11 lab sections and 1 of 12 problem session sections. Any combination of lecture, lab, and problem session is possible. How many different ways can a single student be registered for the physics course?

Solution. To use the multiplication principle, we need to identify the steps of the procedure and the number of outcomes possible for each step. We also must convince ourselves that the process for each step does not depend on the process at any other step. This independence will be obvious. The steps proceed as follows: Choose a section, choose a lab, and choose a problem session. There are 6 possible outcomes for the first step, 11 for the second, and 12 for the third. Therefore, the Multiplication Principle gives

$$\begin{aligned}(\text{\# Ways to register for physics course}) &= (\text{\# Choices for section}) \cdot (\text{\# Choices for lab}) \\ &\quad \cdot (\text{\# Choices for problem session}) \\ &= 6 \cdot 11 \cdot 12 \\ &= 792\end{aligned}$$

Example 2.5. Suppose that Fast Lease, Inc., has three brands of microcomputers available for lease.Each microcomputer is leased together with one of four different software packages. To have leases available for customers to sign regardless of the options they choose, how many different leases should be drawn up and available at any time?

Solution: The solution of this problem uses the following kind of analysis: First, a customer must choose a microcomputer from among the three possibilities. Second, after choosing a microcomputer, the customer must choose an appropriate software package, which can be done in four ways no matter which microcomputer was chosen. Consequently, the total number of choices is

$$\begin{aligned}(\#\text{Choices for microcomputer}) \cdot (\#\text{Choices for software package}) &= 3 \cdot 4 \\ &= 12\end{aligned}$$

Fast Lease should therefore have 12 different leases available at all times.

Example 2.6. Single characters displayed on a digital watch's display are formed by turning on some of the areas in a rectangular grid. Figure 2.1 shows the seven lines that a typical digital watch uses to form characters. How many different characters can be formed?

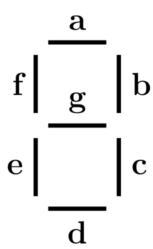


Figure 2.1: Figures on a digital watch's display.

Solution: The digits can be represented as shown in Figure 2.2.

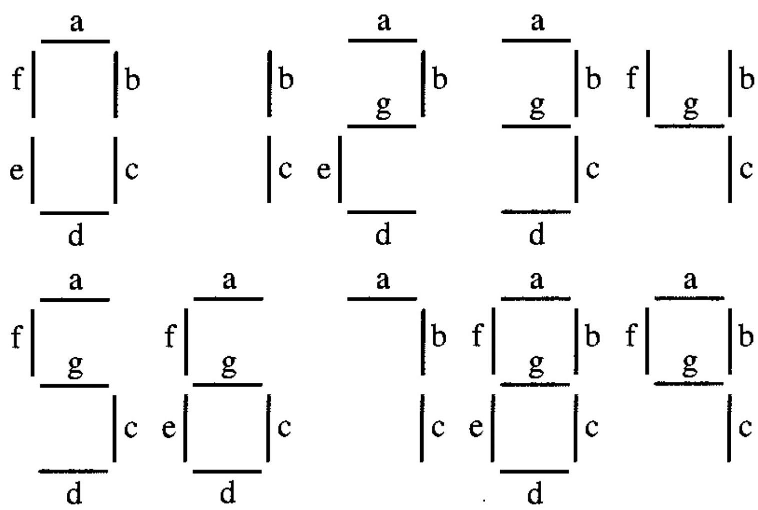


Figure 2.2: Numbers of digital watch

Notice that a total of seven different line segments are used in forming the different numerals. You can form a pattern of lines by indicating for each of the seven line segments whether it is "on" or "off." Since there are seven line segments and each can be on or off independently of whether any other line segment is on or off, the Multiplication Principle says that $2^7 = 128$ characters can be represented by this grid.

Example 2.7. Show that the number of subsets of a finite set with n elements is 2^n.

Solution. Let the elements of an n-element set be $a_1, a_2, \ldots, a_n$. Each subset of these elements can be uniquely associated with an n-tuple of 0's and 1's, with 1 in the ith position indicating that the element a_i is in the subset and 0 in the ith position indicating the element a_i is not in the subset. In this procedure, for $1 \leq i \leq n$, the ith step has one of two possible outcomes: Either the ith element of the set is an element of the subset and the entry is 1, or the ith element of the set is not an element of the subset and the entry is 0. Therefore, by the Multiplication Principle, there are 2^n such n-tuples and, consequently, 2^n possible subsets of an n-element set.

2.2 Principle of Inclusion and Exclusion

The inclusion-exclusion principle is a fundamental rule in combinatorics used to find the number of elements in the union of multiple sets, especially when the sets overlap. It corrects for over-counting by subtracting intersections.
Consider a finite set A containing p number of elements. Here, the number p is called **order**, size or the **cardinality** of the set A and is denoted by $o(A)$, or $n(A)$ or $|A|$.
For example, if $A = \{1, 2, 6\}$ and $B = \{a, b, c, d\}$ then $o(A) = |A| = 3$ and $o(B) = |B| = 4$.

Theorem 2.8 (Basic Counting Theorem)**.** *Let A and B be subsets of a finite universal set U.*

1. *Let $B \subseteq A$. Then:*
   (a) $|B| \leq |A|$
   (b) $|A - B| = |A| - |B|$
   (c) $|B| = |A|$ *if and only if* $B = A$
2. *Let A and B be disjoint finite sets. Then:*
   - $|A \cap B| = 0$
   - $|A \cup B| = |A| + |B|$
3. $|\overline{A}| = |U| - |A|$

Proof. The proof is left to the reader. □

Note 2: F or any two finite sets A and B, if $A \subseteq B$ then $|A| \leq |B|$ and if $A \subset B$ then $|A| < |B|$.

Note 3: If A is a subset of a finite universal set U, then the number of elements in the complement $\overline{A}$ (of A in U) is given by:

$$|\overline{A}| = |U| - |A| \tag{2.1}$$

Theorem 2.9 (Principle of Inclusion-Exclusion for Two Sets)**.** *Let A and B be finite sets. Then:*

$$|A \cup B| = |A| + |B| - |A \cap B| \tag{2.2}$$

Proof. The set $A \cup B$ can be partitioned as:

$$A \cup B = (A - B) \cup (A \cap B) \cup (B - A)$$

where any two of $(A - B)$, $(A \cap B)$, and $(B - A)$ are disjoint (by Theorem 6 in Section 1.3.2). This implies that $(A - B)$ and $(A \cap B) \cup (B - A)$ are also disjoint. Applying Theorem 1(b) gives:

$$\begin{aligned} |A \cup B| &= |A - B| + |(A \cap B) \cup (B - A)| \\ &= |A - B| + |A \cap B| + |B - A| \end{aligned} \tag{2.3}$$

By Theorem 6(b), we have:

$$|A| = |A - B| + |A \cap B| \tag{2.4}$$

$$|B| = |A \cap B| + |B - A| \tag{2.5}$$

Adding equations (2.4) and (2.5) yields:

$$|A| + |B| = |A - B| + 2 \cdot |A \cap B| + |B - A|$$

$$|A| + |B| - |A \cap B| = |A - B| + |A \cap B| + |B - A| \tag{2.6}$$

Substituting equation (2.6) into (2.3) gives the required result:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

□

Example 2.10. Suppose:

$$|A| = 30, \quad |B| = 25, \quad |A \cap B| = 10$$

Then:

$$|A \cup B| = 30 + 25 - 10 = 45$$

Example 2.11. The population of Atlanteas is 830, of which 250 are adult females and 380 are children.

1. How many adults live in Atlanteas?
2. How many adult males live in Atlanteas?
3. How many "females and children" live in Atlanteas?

Solution. The universal set is $U = \{\text{residents of Atlanteas}\}$. The subsets of interest are

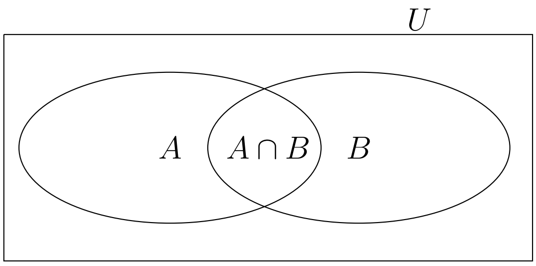
$$A = \{\text{adults}\}, \quad F = \{\text{adult females}\}, \quad M = \{\text{adult males}\}, \quad \text{and } C = \{\text{children}\}$$

1. $|A| = |U| - |C| = 830 - 380 = 450$ (by Theorem 1(c))
2. $|M| = |A| - |F| = 450 - 250 = 200$ (since $F \subseteq A$, using Theorem 1(a.ii))
3. $|F \cup C| = |F| + |C| - |F \cap C| = 250 + 380 - 0 = 630$ (since $F \cap C = \emptyset$, using Theorem 1(b))

Example 2.12. A computer company needs to hire:

- 25 systems programmers (set A)
- 40 applications programmers (set B)
- 10 programmers for both roles ($A \cap B$)

Solution using Inclusion-Exclusion:



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Calculating the total hires:

$$|A \cup B| = 25 + 40 - 10 = 55 \text{ programmers}$$

Remarks

The results in parts 1 and 2 of Theorem 1 are very special, because they make strong assumptions about A and B. If these assumptions fail, the conclusions are generally incorrect.

Example 2.13 (Card Counting Problem)**.** A standard deck of cards has:

- Four suits: Clubs, Diamonds, Hearts, and Spades
- Diamonds and Hearts are red suits; Clubs and Spades are black suits
- Each suit contains 13 cards: Ace (1), 2–10, Jack, Queen, King

How many cards are black or have the value of 3?

Solution: Let:

- A = set of black cards $\Rightarrow |A| = 26$ (13 Clubs + 13 Spades)
- B = set of 3's $\Rightarrow |B| = 4$ (one 3 per suit)
- $A \cap B$ = black 3's $\Rightarrow |A \cap B| = 2$ (3 of Clubs and 3 of Spades)

Applying the Principle of Inclusion-Exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B| = 26 + 4 - 2 = 28$$

Thus, there are $\boxed{28}$ cards that are either black or have the value 3. □

Classwork. A tennis camp has 39 players. There are 25 left-handed players and 22 players who have a two-handed back stroke. How many left-handed players have a two-handed back stroke if every player is represented in these two counts?

Theorem 2.14 (Principle of Inclusion-Exclusion for Three Sets)**.** *Let A, B, and C be finite sets. Then,*

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \tag{2.7}$$

Proof. We proceed by applying Theorem 2.9 (Inclusion-Exclusion for Two Sets) repeatedly:

$$\begin{aligned}
|A \cup B \cup C| &= |(A \cup B) \cup C| \quad \text{(by definition of union)} \\
&= |A \cup B| + |C| - |(A \cup B) \cap C| \quad \text{(by Theorem 2)} \\
&= |A \cup B| + |C| - |(A \cap C) \cup (B \cap C)| \quad \text{(Distributive Law)} \\
&= |A \cup B| + |C| - \big(|A \cap C| + |B \cap C| - |(A \cap C) \cap (B \cap C)|\big) \quad \text{(by Theorem 2)} \\
&= |A \cup B| + |C| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad \text{(simplifying intersection)} \\
&= \big(|A| + |B| - |A \cap B|\big) + |C| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad \text{(applying Th.2 to } |A \cup B \\
&= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad \text{(final simplification)}
\end{aligned}$$

This completes the proof. □

Example 2.15. Let $|A| = 20, \quad |B| = 15, \quad |C| = 10,$

$$|A \cap B| = 5, \quad |B \cap C| = 3, \quad |C \cap A| = 2, \quad |A \cap B \cap C| = 1$$

Then using principle of inclusion-exclusion for three sets

$$|A \cup B \cup C| = 20 + 15 + 10 - 5 - 3 - 2 + 1 = 36$$

Example 2.16. How many natural numbers between 1 and 30,000,000 (including 1 and 30,000,000) are divisible by 2, 3, or 5?

Solution. Let

$$D_i = \{n \in \mathbb{N} : 1 \leq n \leq 30,000,000 \text{ and } n \text{ is divisible by } i\}$$

What is $|D_2 \cup D_3 \cup D_5|$? The number is difficult to count directly, so we use the Principle of Inclusion-Exclusion. $|D_2| = 15,000,000$, $|D_3| = 10,000,000$, and $|D_5| = 6,000,000$. How about $|D_2 \cap D_3|$? Since 2 and 3 are both prime, an integer n is divisible by both 2 and 3 if and only if n is divisible by $2 \cdot 3 = 6$. So $D_2 \cap D_3 = D_6$, and $|D_6| = 5,000,000$. Similarly,

$$|D_2 \cap D_5| = |D_{10}| = 3,000,000$$

$$|D_3 \cap D_5| = |D_{15}| = 2,000,000$$

and

$$|D_2 \cap D_3 \cap D_5| = |D_{30}| = 1,000,000$$

Now, by the Principle of Inclusion-Exclusion for Three Sets,

$$\begin{aligned}
|D_2 \cup D_3 \cup D_5| = |D_2| + |D_3| + |D_5| \\
&- |D_2 \cap D_3| - |D_2 \cap D_5| - |D_3 \cap D_5| \\
&+ |D_2 \cap D_3 \cap D_5| \\
&= 22,000,000
\end{aligned}$$

3. General Case (n Sets)

Let $A_1, A_2, \dots, A_n$ be finite sets. The number of elements in the union is given by:

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i=1}^{n} |A_i| - \sum_{1\le i<j\le n} |A_i \cap A_j| + \sum_{1\le i<j<k\le n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1}|A_1 \cap A_2 \cap \cdots \cap A_n|$$

Application Example: Divisibility Problem

Problem: How many numbers between 1 and 100 are divisible by 2 or 3?
Define:

$$A = \{x \le 100, 2 \mid x\} \quad \Rightarrow \quad \left\lfloor \frac{100}{2} \right\rfloor = 50$$

$$B = \{x \le 100, 3 \mid x\} \quad \Rightarrow \quad \left\lfloor \frac{100}{3} \right\rfloor = 33$$

$$A \cap B = \{x \le 100, 6 \mid x\} \quad \Rightarrow \quad \left\lfloor \frac{100}{6} \right\rfloor = 16$$

Using principle of inclusion-exclusion for two sets

$$|A \cup B| = 50 + 33 - 16 = 67$$

So, there are **67 numbers** divisible by 2 or 3 between 1 and 100.

Application of De Morgan's law

If U is a finite universal set of which A and B are subsets, then, by virtue of a De Morgan's Law and the expression (2.1) above, we have–

$$|\overline{A \cap B}| = |\overline{A} \cup \overline{B}| = |U| - |A \cap B| \tag{2.8}$$

With the use of formula (2.2) above, this becomes

$$\begin{aligned} |\overline{A \cap B}| &= |U| - \{|A| + |B| - |A \cap B|\} \\ &= |U| - |A| - |B| + |A \cap B| \end{aligned} \tag{2.9}$$

Expressions (2.8) and (2.9) are equivalent to one another. Either of these is referred to as the **Addition Principle** (Rule) or the **Principle of inclusion-exclusion** for two sets.
In the particular case where A and B are disjoint sets so that $A \cap B = \emptyset$, the addition rule (2.2) becomes–

$$|A \cup B| = |A| + |B| - |\emptyset| = |A| + |B| \tag{2.10}$$

This is known as the **Principle of disjunctive counting** for two sets.

General principle of disjunctive counting

If a set U is the union of disjoint nonempty subsets $A_1, \ldots, A_n$, then

$$|U| = |A_1| + |A_2| + \ldots + |A_n|.$$

We emphasize that the subsets $A_1, A_2, \ldots, A_n$ must have no elements in common. Moreover, since

$$U = A_1 \cup A_2 \cup \ldots \cup A_n,$$

each element of U is in exactly one of the subsets A_i. In other words, $A_1, A_2, \ldots, A_n$ is a partition of U.

If the subsets $A_1, A_2, \ldots, A_n$ were allowed to overlap, then a more profound principle will be needed—the principle of inclusion and exclusion.

2.2.1 Derangements

Given a set $A = \{a_1, a_2, \ldots, a_n\}$, we know that the number of bijections from A to itself is $n!$. How many such bijections are there that map no element $a \in A$ to itself? That is, how many bijections are there of the form $f : A \to A$, such that $f(a) \neq a$ for all $a \in A$? These are called *derangements*, or bijections with no *fixed points*.

We can reason as follows: Let S_i be the set of bijections that map the i-th element of A to itself. We are then looking for the quantity

$$n! - \left|\bigcup_{i=1}^{n} S_i\right|.$$

By the inclusion-exclusion principle, this is

$$n! - \sum_{k=1}^{n}(-1)^{k-1} \sum_{1\leq i_1<i_2<\cdots<i_k\leq n} |S_{i_1} \cap S_{i_2} \cap \ldots \cap S_{i_k}|.$$

Consider an intersection $S_{i_1} \cap S_{i_2} \cap \ldots \cap S_{i_k}$. Its elements are the permutations that map $a_{i_1}, a_{i_2}, \ldots, a_{i_k}$ to themselves. The number of such permutations is $(n-k)!$, hence

$$|S_{i_1} \cap S_{i_2} \cap \ldots \cap S_{i_k}| = (n-k)!.$$

This allows expressing the number of derangements as:

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

2.2.2 Product Rule: The Principle of Sequencing Counting

If $A_1, \ldots, A_n$ are **nonempty sets**, then the number of elements in the Cartesian product $A_1 \times A_2 \times \cdots \times A_n$ is the product $\prod_{i=1}^{n} |A_i|$. That is,

$$|A_1 \times A_2 \times \cdots \times A_n| = \prod_{i=1}^{n} |A_i|.$$

Particular case for two sets

If there are 5 branches in the first stage corresponding to the 5 elements of A_1 and to each of these branches there are 3 branches in the second stage corresponding to the 3 elements of A_2, then a total of 15 branches altogether.
Moreover, the Cartesian product $A_1 \times A_2$ can be partitioned as $(a_1 \times A_2) \cup (a_2 \times A_2) \cup (a_3 \times A_2) \cup (a_4 \times A_2) \cup (a_5 \times A_2)$, where $(a_i \times A_2) = \{(a_i, b_1), (a_i, b_2), (a_i, b_3)\}$. Thus, for example, $(a_3 \times A_2)$ corresponds to the third branch in the first stage followed by each of the 3 branches in the second stage.

General case for two sets

If $a_1, \ldots, a_n$ are the n distinct elements of A_1 and $b_1, \ldots, b_m$ are the m distinct elements of A_2, then

$$A_1 \times A_2 = \bigcup_{i=1}^{n} (a_i \times A_2).$$

For if x is an arbitrary element of $A_1 \times A_2$, then $x = (a, b)$ where $a \in A_1$ and $b \in A_2$. Thus, $a = a_i$ for some i and $b = b_j$ for some j. Thus, $x = (a_i, b_j) \in (a_i \times A_2)$ and therefore $x \in \bigcup_{i=1}^{n} (a_i \times A_2)$.
Conversely, if $x \in \bigcup_{i=1}^{n} (a_i \times A_2)$, then $x \in (a_i \times A_2)$ for some i and thus $x = (a_i, b_j)$ where b_j is some element of A_2. Therefore, $x \in A_1 \times A_2$.
Next observe that $(a_i \times A_2)$ and $(a_j \times A_2)$ are disjoint if $i \neq j$ since if
$x \in (a_i \times A_2) \cap (a_j \times A_2)$ then $x = (a_i, b_k)$ for some k and $x = (a_j, b_l)$ for some l.
But then $(a_i, b_k) = (a_j, b_l)$ implies that $a_i = a_j$ and $b_k = b_l$. But since $i \neq j$, $a_i \neq a_j$.
Thus, we conclude that $A_1 \times A_2$ is the disjoint union of the sets $(a_i \times A_2)$. Furthermore $|a_i \times A_2| = |A_2|$ since there is obviously a one-to-one correspondence between the sets $a_i \times S_2$ and A_2, namely, $(a_i, b_j) \to b_j$.
Then by the sum rule

$$|A_1 \times A_2| = \sum_{i=1}^{n} |a_i \times A_2| = |A_2| + |A_2| + \ldots + |A_2| = n|A_2| = mn$$

Therefore, we have proven the product rule for two sets. The general rule follows by mathematical induction.

2.3 Permutation and Combinations

2.3.1 Permutation

A **permutation** of n objects taken r at a time (also called an r-permutation of n objects) is an ordered selection or arrangement of r of the objects.
Note: We are simply defining the terms r-combinations and r-permutations here and have not mentioned anything about the properties of the n objects.
For example, these definitions say nothing about whether or not a given element may appear more than once in the list of n objects.
In other words, it may be that the n objects do not constitute a set in the normal usage of the word.

General formulas for enumerating combinations and permutations will now be presented. At this time, we will only list formulas for combinations and permutations without repetitions or with unlimited repetitions. We will wait until later to use generating functions to give general techniques for enumerating combinations where other rules govern the selections.
Let $P(n, r)$ denote the number of r-permutations of n elements without repetitions.

Problem

How many different sequences, each of length r, can be formed using elements from A if
(a) elements in the sequence may be repeated?
(b) all elements in the sequence must be distinct?

Solution: First, we note that any sequence of length r can be formed by filling in r boxes in order from left to right with elements of A. In case (a) we may use copies of elements of A.

Box 1	Box 2	$\cdots$	Box $r-1$	Box r

Let T_1 be the task "fill box 1," let T_2 be the task "fill box 2," and so on. Then the combined task $T_1T_2\cdots T_r$ represents the formation of the sequence.

Case (a). T_1 can be accomplished in n ways, since we may copy any element of A for the first position of the sequence. The same is true for each of the tasks $T_2, T_3, \ldots, T_r$. Then, by the extended multiplication principle, the number of sequences that can be formed is

$$\underbrace{n \cdot n \cdot \cdots \cdot n}_{r \text{ factors}} = n^r.$$

We have therefore proved the following result.

Theorem 2.17. *Let A be a set with n elements and $1 \le r \le n$. Then the number of sequences of length r that can be formed from elements of A, allowing repetitions, is n^r.*

Example 2.18. How many three-letter "words" can be formed from letters in the set $\{a, b, y, z\}$ if repeated letters are allowed?

Solution: Here n is 4 and r is 3, so the number of such words is 4^3 or 64, by Theorem 3.
Now we consider case (b) of Problem 1. Here, also, T_1 can be performed in n ways, since any element of A can be chosen for the first position. Whichever element is chosen, only $(n-1)$ elements remain, so T_2 can be performed in $(n-1)$

Case (b).

Theorem 2.19 (Enumerating r-permutations without repetitions)**.**

$$P(n, r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

Proof. Since there are n distinct objects, the first position of an r-permutation may be filled in n ways. This done, the second position can be filled in $n-1$ ways since no repetitions are allowed and there are $n-1$ objects left to choose from. The third can be filled in $n-2$ ways. By applying the product rule, we conclude that

$$P(n,r) = n(n-1)(n-2)\cdots(n-r+1).$$

From the definition of factorials, it follows that

$$P(n,r) = \frac{n!}{(n-r)!}$$

□

Example 2.20. Let A be $\{1,2,3,4\}$. Then the sequences 124, 421, 341, and 243 are some permutations of A taken 3 at a time. The sequences 12, 43, 31, 24, and 21 are examples of different permutations of A taken two at a time.
By Theorem 4, the total number of permutations of A taken three at a time is

$$P(4,3) = 4\cdot 3\cdot 2 = 24$$

. The total number of permutations of A taken two at a time is

$$P(4,2) = 4\cdot 3 = 12.$$

Example 2.21 (Traveling Salesman Problem)**.** A more general description of a solution for n cities gives insight regarding the number of operations required to find a solution. Assume that n cities need to be visited and that one city is designated as the starting point for all routes. There are $n-1$ cities that remain to be visited immediately after the starting city. Thus, there are $n-1$ ways to choose the second city to visit. For each of these $n-1$ partial routes consisting of city 1 and the choice of a second city, there are $n-2$ remaining unvisited cities. In all, we have $(n-1)\cdots(n-i)$ partial routes starting at city 1 and containing $i+1$ cities. The total number of routes starting at one city and not returning to that city until all the other cities have been visited is $(n-1)\cdots(n-(n-1)) = (n-1)!$. Therefore, $(n-1)!$ routes exist for visiting each of the n cities exactly once provided that one city is designated as the starting and ending city.
For instance, find a best route for visiting the four cities with the travel times given.

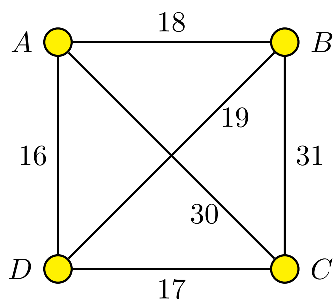


Figure 2.3: Four-city network with travel time

Note: When $r = n$, this formula becomes

$$P(n,n) = \frac{n!}{0!} = n!$$

When no explicit reference to r is made, we assume that all the objects are to be arranged; thus when we talk about the permutations of n objects we mean the case $r = n$.

Corollary 2.22. *There are $n!$ permutations of n distinct objects.*

Number of permutations that can be formed from a collection of n objects of which n_1 are of one type, n_2 are of second type $\ldots$, n_k are of k^{th} type with $n_1 + n_2 + \cdots + n_k = n$. Then the number of permutations of the n objects is

$$\frac{n!}{n_1! n_2! \cdots n_k!}$$

Example 2.23. There are $3! = 6$ permutations of $\{a, b, c\}$.
There are $4! = 24$ permutations of $\{a, b, c, d\}$.
The number of 2-permutations of $\{a, b, c, d, e\}$ is $P(5, 2) = \frac{5!}{(5-2)!} = 5 \times 4 = 20$.
The number of 5-letter words using the letters a, b, c, d, and e at most once is $P(5, 5) = 120$.

Example 2.24. There are $P(10, 4) = 5{,}040$ 4-digit numbers that contain no repeated digits since each such number is just an arrangement of four of the digits $0, 1, 2, 3, \ldots, 9$ (leading zeroes are allowed). There are $P(26, 3) \times P(10, 4)$ license plates formed by 3 distinct letters followed by 4 distinct digits.

Example 2.25. In how many ways can 7 women and 3 men be arranged in a row if the 3 men must always stand next to each other?

Solution: There are 3! ways of arranging the 3 men. Since the 3 men always stand next to each other, we treat them as a single entity, which we denote by X. Then if $W_1, W_2, \ldots, W_7$ represent the women, we next are interested in the number of ways of arranging $\{X, W_1, W_2, W_3, \ldots, W_7\}$. There are 8! permutations of these 8 objects. Hence there are (3!)(8!) permutations altogether. (Of course, if there has to be a prescribed order of arrangement for the 3 men, then there are only 8! total permutations).

Example 2.26. In how many ways can the letters of the English alphabet be arranged so that there are exactly 5 letters between the letters a and b?

Solution: There are $P(24, 5)$ ways to arrange the 5 letters between a and b, 2 ways to place a and b, and then 20! ways to arrange any 7-letter word treated as one unit along with the remaining 19 letters. The total is $P(24, 5) \times 20! \times 2$.
Note: If instead of arranging objects in a line, we arrange them in a circle, then the number of permutations decreases.

Example 2.27. In how many ways can 5 children arrange themselves in a ring?

Solution: Here, the 5 children are not assigned to particular places but are only arranged relative to one another. Thus, the arrangements are considered the same if the children are in the same order clockwise. Hence, the position of child C_1 is immaterial and it is only the position of the 4 other children relative to C_1 that counts. Therefore, keeping C_1 fixed in position, there are 4! arrangements of the remaining children.

Some worked out examples

1. *How many different strings (sequences) of length 4 can be formed using the letters of the word FLOWER.*

   Sol: The given word FLOWER has 6 letters where all of which are distinct. The required number of strings is
   $$P(6,4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = 360$$

2. *Find the number of permutations of the letters of the word SUCCESS.*

   Sol: The given word SUCCESS has 7 letters, of which 3 are S's, 2 are C's and 1 each are U and E. The required number of permutations is
   $$\frac{7!}{3!\,2!\,1!\,1!} = 420$$

3. *Find the number of permutations of the letters of the word MASSASAUGA. In how many of these, all four A's are together? How many of them begin with S?*

   Sol: The given word MASSASAUGA has 10 letters of which 4 are A's, 3 are S's and 1 each are M, U and G. Required number of permutations is
   $$\frac{10!}{4!\,3!\,1!\,1!\,1!} = 25{,}200$$

   If all A's are together, we treat all A's as one single letter.

   Then required number of permutations is
   $$\frac{7!}{1!\,3!\,1!\,1!\,1!} = 840$$

   If the word begins with letter S, there occurs 9 open positions to fill, where 2 are S's, 4 are A's and one each are M, U, G.

   Then required number of permutations is
   $$\frac{9!}{2!\,4!\,1!\,1!\,1!\,1!} = 7{,}560$$

4. *How many positive integers n can we form using the digits 3,4,4,5,5,6,7 if we want n to exceed 5,000,000?*

   Sol: Here n must be of the form $n = x_1x_2x_3x_4x_5x_6x_7$ where $x_1, x_2, \ldots, x_7$ are the given digits with $x_1 = 5$ or 6 or 7.

   - Suppose we take $x_1 = 5$ then $x_2, \ldots, x_7$ is an arrangement of the remaining 6 digits which contains two 4's and one each of 3,5,6,7.

     The number of such arrangements is
     $$\frac{6!}{2!\,1!\,1!\,1!\,1!} = 360$$

- Next, suppose we take $x_1 = 6$ then $x_2, \ldots, x_7$ is an arrangement of the remaining 6 digits which contains two each of 4 and 5 and one each of 3,7.

  The number of such arrangements is

$$\frac{6!}{2!\,2!\,1!\,1!} = 180$$

- Similarly, we take $x_1 = 7$ then $x_2, \ldots, x_7$ is an arrangement of the remaining 6 digits which contains two each of 4 and 5 and one each of 3,6.

  The number of such arrangements is

$$\frac{6!}{2!\,2!\,1!\,1!} = 180$$

According to Sum Rule, the number of n's of desired type is

$$360 + 180 + 180 = 720$$

5. *Four different Mathematics books, five different Computer Science books and two different Control Theory books are to be arranged in a shelf. How many different arrangements are possible if*

   (a) the books in each particular subject must all be together?

   (b) Only the Mathematics books must be together?

   Sol:

   (a) The Mathematics books can be arranged among themselves in 4! ways, the Computer Science books in 5! ways, the Control Theory books in 2! ways, and the three groups in 3! ways.

   The number of possible arrangements is

$$4! \times 5! \times 2! \times 3! = 34{,}560$$

   (b) Consider the four Mathematics books as one single book. Then we have 8 books which can be arranged in 8! ways. In all of these ways the Mathematics books are together. But the Mathematics books can be arranged among themselves in 4! ways. Hence, the number of arrangements is

$$8! \times 4! = 967{,}680$$

6. *Find the value of n such that $2P(n,2) + 50 = P(2n,2)$.*

   Sol:

$$\begin{aligned}
2P(n,2) + 50 &= P(2n,2) \\
2 \times \frac{n!}{(n-2)!} + 50 &= \frac{(2n)!}{(2n-2)!} \\
2n(n-1) + 50 &= 2n(2n-1) \\
n^2 &= 25 \\
n &= 5 \text{ or } -5
\end{aligned}$$

Since n cannot be negative, the value of $n = 5$.

2.3.2 Combinations

The multiplication principle and the counting methods for permutations all apply to situations where order matters. In this section we look at some counting problems where order does not matter.

Problem. Let A be any set with n elements and $1 \leq r \leq n$. How many different subsets of A are there, each with r elements?
The traditional name for an r-element subset of an n-element set A is a **combination of A, taken r at a time.**

Example 2.28. Let $A = \{1, 2, 3, 4\}$. The following are all distinct combinations of A, taken three at a time:

$$A_1 = \{1, 2, 3\}, \quad A_2 = \{1, 2, 4\}, \quad A_3 = \{1, 3, 4\}, \text{ and } A_4 = \{2, 3, 4\}.$$

Note that these are subsets, not sequences. Therefore,

$$A_1 = \{2, 1, 3\} = \{2, 3, 1\} = \{1, 3, 2\} = \{3, 1, 2\} = \{3, 2, 1\}.$$

In other words, when it comes to combinations, unlike permutations, the order of the elements is irrelevant.

Example 2.29. Let A be the set of all 52 cards in an ordinary deck of playing cards. Then a combination of A, taken five at a time, is just a hand of five cards regardless of how these cards were dealt. ♦

We now want to count the number of r-element subsets of an n-element set A. This is most easily accomplished by using what we already know about permutations. Observe that each permutation of the elements of A, taken r at a time, can be produced by performing the following two tasks in sequence.
Task 1. Choose a subset B of A containing r elements.
Task 2. Choose a particular permutation of B.
We are trying to compute the number of ways to choose B. Call this number C. Then task 1 can be performed in C ways, and task 2 can be performed in $r!$ ways. Thus the total number of ways of performing both tasks is, by the multiplication principle, $C \cdot r!$. But it is also $P(n, r)$. Hence

$$C \cdot r! = P(n, r) = \frac{n!}{(n-r)!}.$$

Therefore,

$$C = \frac{n!}{r!(n-r)!}.$$

We have proved the following result.

Theorem 2.30. *Let A be a set with $|A| = n$, and let $1 \leq r \leq n$. Then the number of combinations of the elements of A, taken r at a time, that is, the number of r-element subsets of A, is*

$$\frac{n!}{r!(n-r)!}.$$

Note again that the number of combinations of A, taken r at a time, does not depend on A, but only on n and r. This number is often written $C(n,r)$ and is called the **number of combinations of n objects taken r at a time**. We have

$$C(n,r) = \frac{n!}{r!(n-r)!} = \binom{n}{r}.$$

This computation can be done directly on many calculators.

Example 2.31. Compute the number of distinct five-card hands that can be dealt from a deck of 52 cards.

Solution: This number is $C(52,5) = \frac{52!}{5!\,47!} = 2,598,960$, because the order in which the cards were dealt is irrelevant. Compare this number with the number computed in Section 3.1, Example 6.

Theorem 2.32. *Suppose that k selections are to be made from n items without regard to order and that repeats are allowed, assuming at least k copies of each of the n items. The number of ways these selections can be made is*

$$\binom{n+k-1}{k}.$$

Example 2.33. In how many ways can a prize winner choose three CDs from the Top Ten list if repeats are allowed?

Solution: Here $n = 10$ and $k = 3$. By Theorem 2, there are $C(10+3-1) = C(12,3)$ ways to make the selections. The prize winner can make the selection in 220 ways.

Example 2.34. A certain question paper contains 3 parts A, B, C with 4 questions in part A, 5 questions in part B and 6 questions in part C. It is required to answer 7 questions selecting at least 2 questions from each part. In how many different ways can a student select his seven questions for answering?

Solution: The different possible ways in which a student can make a selection are:

(1) 2 questions from part A, 2 from part B and 3 from part C

(2) 2 questions from part A, 3 from part B and 2 from part C

(3) 3 questions from part A, 2 from part B and 2 from part C

- Selection (1) can be made in $C(4,2) \times C(5,2) \times C(6,3) = 1200$ ways
- Selection (2) can be made in $C(4,2) \times C(5,3) \times C(6,2) = 900$ ways
- Selection (3) can be made in $C(4,3) \times C(5,2) \times C(6,2) = 600$ ways

Therefore, the number of possible selections is $1200 + 900 + 600 = 2700$.

Note: In general, when order matters, we count the number of sequences or permutations; when order does not matter, we count the number of subsets or combinations. Some problems require that the counting of permutations and combinations be combined or supplemented by the direct use of the addition or the multiplication principle.

Example 2.35. Suppose that a valid computer password consists of seven characters, the first of which is a letter chosen from the set $\{A, B, C, D, E, F, G\}$, and the remaining six characters are letters chosen from the English alphabet or a digit. How many different passwords are possible?

Solution: A password can be constructed by performing the tasks T_1 and T_2 in sequence.
Task T_1: Choose a starting letter from the set given.
Task T_2: Choose a sequence of letters and digits. Repeats are allowed.
Task T_1 can be performed in $C_1 = 7$ ways. Since there are 26 letters and 10 digits that can be chosen for each of the remaining six characters, and since repeats are allowed, task T_2 can be performed in $36^6 = 2{,}176{,}782{,}336$ ways.
By the multiplication principle, there are $7 \times 2{,}176{,}782{,}336 = 15{,}237{,}476{,}352$ different passwords.

Example 2.36. How many different seven-person committees can be formed each containing 3 women from an available set of 20 women and 4 men from an available set of 30 men?

Solution: In this case a committee can be formed by performing the following two tasks in succession:
Task 1: Choose 3 women from the set of 20 women.
Task 2: Choose 4 men from the set of 30 men.
Here order does not matter in the individual choices, so we are merely counting the number of possible subsets. Thus task 1 can be performed in $C(20, 3) = 1{,}140$ ways and task 2 can be performed in $C(20, 3) = 27{,}405$ ways.
By the multiplication principle, there are $(1{,}140)(27{,}405) = 31{,}241{,}700$ different committees.

Properties of permutation and combinations

1. $P(n, r) = \dfrac{n!}{(n-r)!}$
2. $C(n, r) = \dfrac{n!}{(n-r)!r!}$
3. $P(n, 0) = 1$
4. $P(n, n) = P(n, n-1) = n!$
5. $P(n, r) = nP(n-1, r-1)$
6. $\dfrac{P(n, r)}{P(n, r-1)} = n - r + 1$
7. $P(n-1, r) + rP(n-1, r-1) = P(n, r)$
8. $P(n, r) = r! \cdot C(n, r)$
9. $C(n, 0) = C(n, n) = 1$
10. $C(n, 1) = C(n, n-1) = n$
11. $C(n, r) = C(n, n-r)$
12. $C(n, r) + C(n, r-1) = C(n+1, r)$

Exercise

1. There are three cities (A, B, and C) in a country. If there are 4 routes from city A to city B and 5 routes from city B to city C, in how many ways can a person make a tour from city A to

city C through B? If the person is not allowed to use the same routes while returning, in how many ways can the person return from city C to city A? (Ans. 20, 12)

2. In how many ways can you color the boxes $\boxed{}\boxed{}\boxed{}\boxed{}$ with seven different colors? (Ans. 840)

3. How many odd numbers of 5 digits can be formed from the number 2, 0, 4, 3, 8 when repetition of digit is not allowed? (Ans. 18)

4. How many license plates consisting of three digits can be made using $1, 2, 3, 4, 5$? (Ans. 625)

5. How many two-digit even numbers can be formed from the digits 2, 3, 4, 5 and 6 when (i) repetition of digits is not allowed (ii) repetition of digits is allowed? (Ans. 12, 15)

6. In how many ways can 7 persons be arranged in a line such that (a) two particular persons are always together (b) two particular persons are never together. (Ans. 1440, 3600)

7. In how many ways can the letters of the word 'FRIDAY' be arranged? How many of these arrangements do not begin with F? How many of them begin with F and do not end with Y? (Ans. 720, 600, 96)

8. Find the permutations of the letters in the word 'NEPAL'. How many of them (a) begin with N? (b) do not begin with N? (Ans. 120, 24, 96)

9. A lock has 4 digits code to open from the set $\{1, 2, 3, 4, 5, 6\}$. (a) Determine the maximum number of attempts required to open the lock (b) If the code is four different digits odd number, find the maximum number of attempts to open the lock. (Ans. 1296, 180)

10. In how many ways can the letters of the word 'LOGIC' be arranged so that (a) the vowels may occupy odd position? (b) no two vowels are together? (c) the relative positions of the vowels and consonants are not changed? (Ans. 36, 72, 12)

11. In how many ways can the letters of the following words be arranged?
(a) MATHEMATICS (b) MISSISSIPPI (c) PATHSHALA (Ans. 840, $\frac{11!}{4!4!2!}$, $\frac{9!}{3!2!}$)

12. In how many ways can the letters of the word IMPOSSIBLE be arranged so that all the vowels come together? (Ans. 30240)

13. How many different license plates can be made using 3 letters from (A-Z) followed by 3 digits from (0-9)? (Ans. 15,818,400)

14. In how many ways can 4 boys and 4 girls be arranged alternatively at a round table? (Ans. 144)

15. Apsara is inviting 18 friends for a meeting. In how many ways can they be seated at a round table? In how many ways will two particular friends be seated on either side of Apsara? (Ans. 18!, $2 \cdot 16!$)

16. A disc jockey (DJ) has to choose three songs for the last few minutes of his evening show. There are nine songs that he feels are appropriate for that time slot. In how many ways can he choose and arrange to play three of those nine songs? (Ans.504)

17. How many different ways can a president, vice-president, and treasurer be selected from a group of 10 students? (Ans.720)

Combination

1. If ${}^nC_8 = {}^n C_4$, find the value of n. (Ans. 12)

2. If ${}^nC_{r-1} = 45, {}^n C_r = 120$ and ${}^nC_{r+1} = 210$, find n and r. (Ans. 10, 3)

3. If $C(18, r) = C(18, r + 2)$, find r. (Ans. 8)

4. If $C(20, r + 5) = C(20, 2r - 7)$, find $C(15, r)$. (Ans. 455)

5. Find the value of ${}^rC_4$ if ${}^{16}C_r = {}^{16} C_{r+2}$. (Ans. 35)

6. If ${}^nP_r = 336$ and ${}^nC_r = 56$, find n and r. (Ans.8, 3)

7. If there are 12 persons in a party, and each two of them shake hands with each other, how many hand shakes happen in the party? (Ans. 66)

8. If a committee of 5 is to be formed out of 6 gents and 4 ladies, in how many ways this can be done when (a) at least two ladies are included? (b) at most two ladies are included? (Ans. 186, 186)

9. A man has 7 relatives, 4 of whom are ladies; his wife has 7 relatives, 3 of whom are ladies and 4 gentlemen. In how many ways can they invite a dinner party of 3 ladies and 3 gentlemen so that there are 3 of the man's relatives and 3 of the wife's relatives? (Ans. 485)

10. A committee of 5 is to be formed out of 6 gents and 4 ladies. In how many ways this can be done when: (a) at least two ladies are included? (b) at most two ladies are included? (Ans. 186, 186)

11. An examination paper consisting of 10 questions, is divided into two groups A and B. Group A contains 6 questions. In how many ways can an examinee attempt 7 questions (a) selecting 4 from group A and 3 from group B? (b) selecting at least two questions from each group? (Ans. 60, 116)

12. A notice to form a 5-member team to develop "A Students' Notebook" is announced by Pathshala. If 8 girls and 5 boys are equally interested in being on the team, in how many ways can a team be formed with at least 2 girls and at least 1 boy? (Ans. ??)

13. A bookshelf contains 8 fiction books and 5 non-fiction books. In how many ways can 4 books be chosen if exactly 2 must be fiction and 2 must be non-fiction? (Ans. 280)

14. There are 4 balls of color red, green, yellow and blue. In how many ways 2 two balls can be selected such that one of them is red or blue? (Ans. 4)

15. Find the number of subsets of the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$ having 4 elements. (Ans. 330)

Extra questions

1. Prove that the total number of permutations of n different elements taken r at a time is given by
$$^nP_r = \frac{n!}{(n-r)!} \qquad (0 \leq r \leq n)$$
2. Prove that the total number of circular permutations of n different objects is $(n-1)!$.
3. Cricket Association of Nepal (CAN) has 16 cricket players consisting 2 wicket-keepers and 5 bowers. In how many ways can the CAN select a cricket team of eleven players if one wicket-keeper and at least four bowers have to be selected from the group? (Ans. 1092)
4. In how many ways can three letters of the word BIOLOGY be selected? (Ans. 25)
5. Give examples to distinguish row permutation and circular permutation. Also, connect your example for circular permutation considering clockwise and anti-clockwise same.
6. There are three dices each of them having faces with a number from 1 to 6. These dices are rolled. Find the number of possible outcomes such that at least one of the dice shows the number 2. (Ans. 91)
7. Find the number of ways of drawing 9 balls from a bag that has 6 red balls, 5 green balls, and 7 blue balls so that 3 balls of every color are drawn.
8. There are 20 straight lines in a plane so that no two lines are parallel and no three lines are concurrent. Determine the number of points of intersection.
9. There are 3 wicketkeepers are 5 bowlers among 22 cricket players. A team of 11 players is to be selected so that there is exactly one wicketkeeper and at least 4 bowlers in the team. How many teams can be formed?
10. Five students are selected from 11. How many ways can these students be selected if two specified students are not selected?
11. A student finds 7 books of his/her interest, but can borrow only three. Find the number of ways he can choose three books that he wants to borrow.
12. Find the number of ways of dividing 20 objects in three groups of sizes 8, 7 and 5.
13. There are 8 doctors and 4 lawyers in a panel. Find the number of ways for selecting a team of 6 if at least one doctor must be in the team.
14. Four parallel lines intersect another set of five parallel lines. Find the number of distinct parallelograms that can be formed.
15. Find the number of ways in which we can choose a committee from men and six women so that the committee includes at least two men and exactly twice as many women as men.
16. A committee of 6 is to be chosen from 10 men and 7 women so as to contain at least 3 men and 2 women. In how many different ways can this be done if two particular women refuse to serve on the same committee?

17. If a convex polygon has 44 diagonals, find the number of its sides.

18. How many committee of five persons with a chairperson can be selected from 12 persons.

19. A question paper has 6 questions. How many ways does a student have to answer if he wants to solve at least one question?

20. If $C(n+1,3) = 2C(n,2)$, find n.

21. If $P(6,2) = nC(6,2)$, find n.

22. If $C(2n,3) : C(n,2) = 52 : 3$, find n.

23. If $P(n,r) = 720$ and $C(n,n-r) = 120$, find n and r.

24. If $C(2n,3) : C(n,2) = 44 : 3$, find n.

25. If $C(24,x) = C(24,2x+3)$, find x.

26. If $(n+1)! = 12(n-1)!$, find n.

27. Find the total number of ways the letters in KATHMANDU can be arranged. In how many of them is the vowel letter position unchanged?

28. In how many ways can a team of 3 boys and 3 girls be selected from 5 boys and 4 girls?

29. In an examination, a student has to answer 4 questions out of 5 questions; questions 1 and 2 are however compulsory. Determine the number of ways in which the student can make the choice.

30. In how many ways can a student choose 5 courses out of 9 courses if 2 courses are compulsory for every student?

31. How many different five-digit numbered license plates can be made if first digit cannot be zero and the repetition of digits is not allowed?

32. A person wants to buy one fountain pen, one ball pen and one pencil from a stationery shop. If there are 10 fountain pen varieties, 12 ball pen varieties and 5 pencil varieties, in how many ways can he select these articles?

33. How many words (with or without meaning) can be formed using all the letters of the word EQUATION at a time so that the vowels and consonants occur together?

34. There are 15 players in a cricket team, out of which 6 are bowlers, 7 are batsmen and 2 are wicketkeepers. In how many ways can a team of 11 players be selected from them so as to include at least 4 bowlers, 5 batsmen and 1 wicketkeeper?

2.3.3 Pigeonhole Principle (counting method)

In this section we introduce another proof technique, one that makes use of the counting methods we have discussed.

Theorem 2.37 (The Pigeonhole Principle)**.** *If n pigeons are assigned to m pigeonholes, and $m < n$, then at least one pigeonhole contains two or more pigeons.*

Proof. Consider labeling the m pigeonholes with the numbers 1 through m and the n pigeons with the numbers 1 through n. Now, beginning with pigeon 1, assign each pigeon in order to the pigeonhole with the same number. This assigns as many pigeons as possible to individual pigeonholes, but because $m < n$, there are $n - m$ pigeons that have not yet been assigned to a pigeonhole. At least one pigeonhole will be assigned a second pigeon. □

Example 2.38. Show that if any five numbers from 1 to 8 are chosen, then two of them will add up to 9.

Solution: Construct four different sets, each containing two numbers that add up to 9 as follows:

$$A_1 = \{1, 8\}, \quad A_2 = \{2, 7\}, \quad A_3 = \{3, 6\}, \quad A_4 = \{4, 5\}.$$

Each of the five numbers chosen must belong to one of these sets. Since there are only four sets, the pigeonhole principle tells us that two of the chosen numbers belong to the same set. These numbers add up to 9.

Example 2.39. Show that if any 11 numbers are chosen from the set $\{1, 2, \ldots, 20\}$, then one of them will be a multiple of another.

Solution: Every positive integer n can be written as $n = 2^k m$, where m is odd and $k \geq 0$. This can be seen by simply factoring all powers of 2 (if any) out of n. In this case let us call m the *odd part* of n.
If 11 numbers are chosen from the set $\{1, 2, \ldots, 20\}$, then two of them must have the same odd part. This follows from the pigeonhole principle since there are 11 numbers (pigeons), but only 10 odd numbers between 1 and 20 (pigeonholes) that can be odd parts of these numbers.
Let n_1 and n_2 be two chosen numbers with the same odd part. We must have $n_1 = 2^{k_1} m$ and $n_2 = 2^{k_2} m$, for some k_1 and k_2. If $k_1 \geq k_2$, then n_1 is a multiple of n_2; otherwise, n_2 is a multiple of n_1.

Theorem 2.40 (The Extended Pigeonhole Principle)**.** *If n pigeons are assigned to m pigeonholes, then one of the pigeonholes must contain at least $\lfloor (n-1)/m \rfloor + 1$ pigeons.*

Proof by contradiction. If each pigeonhole contains no more than $\lfloor (n-1)/m \rfloor$ pigeons, then there are at most $m \cdot \lfloor (n-1)/m \rfloor \leq m \cdot (n-1)/m = n-1$ pigeons in all. This contradicts our assumptions, so one of the pigeonholes must contain at least $\lfloor (n-1)/m \rfloor + 1$ pigeons. □

Example 2.41. We give an extension of Example 1. Show that if any 30 people are selected, then we may choose a subset of 5 so that all 5 were born on the same day of the week.

Solution: Assign each person to the day of the week on which she or he was born. Then 30 pigeons are being assigned to 7 pigeonholes. By the extended pigeonhole principle with $n = 30$ and $m = 7$, at least $\lfloor (30 - 1)/7 \rfloor + 1 = 5$ of the people must have been born on the same day of the week.

Example 2.42. Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionaries must have at least 2045 pages.

Solution: Let the pages be the pigeons and the dictionaries the pigeonholes. Assign each page to the dictionary in which it appears. Then, by the extended pigeonhole principle, one dictionary must contain at least $\lfloor 61{,}326/30 \rfloor + 1 = 2{,}045$ pages.

2.4 Integers and Divisibility, Greatest Common Divisors, The Division Algorithm for Integers(Without Proof)

2.4.1 Division in the Integers

If m and n are nonnegative integers and n is not zero, we can plot the nonnegative integer multiples of n on a half-line and locate m as in Figure 2.4.

- If m is a multiple of n, say $m = qn$, then we can write $m = qn + r$, where r is 0.
- If m is not a multiple of n, we let qn be the first multiple of n lying to the left of m and let r be $m - qn$. Then r is the distance from qn to m, so clearly $0 < r < n$, and again we have $m = qn + r$ (see Figure 2.4).

Figure 2.4: Number line representation of division algorithm

Theorem 2.43. *If $n \neq 0$ and m are nonnegative integers, we can write $m = qn + r$ for some nonnegative integers q and r with $0 \leq r < n$. Moreover, there is just one way to do this.*

Example 2.44.

(a) If $n = 3$ and $m = 16$, then $16 = 5 \times 3 + 1$, so $q = 5$ and $r = 1$.

(b) If $n = 10$ and $m = 3$, then $3 = 0 \times 10 + 3$, so $q = 0$ and $r = 3$.

- If the r in Theorem 2.43 is zero, so that m is a multiple of n, we write $n \mid m$, which is read "n divides m."
- If m is not a multiple of n, we write $n \nmid m$, which is read "n does not divide m."

Theorem 2.45 (Properties of Divisibility)**.** *Let $a, b,$ and c be integers.*

(a) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.

(b) If $a \mid b$ and $a \mid c$, where $b > c$, then $a \mid (b - c)$.

(c) If $a \mid b$ or $a \mid c$, then $a \mid bc$.

(d) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof. (a) If $a \mid b$ and $a \mid c$, then $b = k_1 a$ and $c = k_2 a$ for some integers k_1 and k_2. So $b + c = (k_1 + k_2)a$ and $a \mid (b + c)$.

(b) This can be proved in exactly the same way as part (a).

(c) As in part (a), we have $b = k_1 a$ or $c = k_2 a$. Then either $bc = k_1 ac$ or $bc = k_2 ab$, so in either case bc is a multiple of a and $a \mid bc$.

(d) If $a \mid b$ and $b \mid c$, we have $b = k_1 a$ and $c = k_2 b$, so $c = k_2 b = k_2(k_1 a) = (k_2 k_1)a$ and hence $a \mid c$.

□

Definition 2.46 (Prime Number)**.** A number $p > 1$ in $\mathbb{Z}^+$ is called **prime** if the only positive integers that divide p are p and 1.

Example 2.47. The numbers 2, 3, 5, 7, 11, and 13 are prime, while 4, 10, 16, and 21 are not prime.

Algorithm to test whether an integer $n > 1$ is prime:

Algorithm 1 (Primality Test for Prime Number)

1: Check whether $n = 2$. If so, n is prime. If not, proceed to
2: Check whether $2 \mid n$. If so, n is not prime; otherwise, proceed to
3: Compute the smallest integer $k \leq \sqrt{n}$. Then
4: Check whether $d \mid n$, where d is any odd number such that $1 < d \leq k$. If $d \mid n$, then n is not prime; otherwise, n is prime.

Theorem 2.48 (Fundamental Theorem of Arithmetic)**.** *Every positive integer $n > 1$ can be written uniquely as*

$$n = p_1^{k_1} p_2^{k_2} \cdots p_s^{k_s},$$

where $p_1 < p_2 < \cdots < p_s$ are distinct primes that divide n and the k_i are positive integers giving the number of times each prime occurs as a factor of n.

Example 2.49. (a) $9 = 3 \cdot 3 = 3^2$

(b) $24 = 8 \cdot 3 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$

(c) $30 = 2 \cdot 3 \cdot 5$

2.4.2 Greatest Common Divisor

✣ If a, b, and k are in $\mathbb{Z}^+$, and $k \mid a$ and $k \mid b$, we say that k is a **common divisor** of a and b.
✣ If d is the largest such k, d is called the **greatest common divisor**, or gcd, of a and b, and we write $d = \gcd(a, b)$.

Theorem 2.50 (Properties of gcd)**.** *If d is $\gcd(a, b)$, then*

(a) $d = sa + tb$ for some integers s and t (these are not both positive).

(b) If c is any other common divisor of a and b, then $c \mid d$.

Example 2.51. (a) The common divisors of 12 and 30 are $2, 3$, and 6, so $\gcd(12, 30) = 6$ and

$$6 = 1 \cdot 30 + (-2) \cdot 12.$$

(b) It is clear that $\gcd(17, 95) = 1$ since 17 is prime and $17 \nmid 95$, and the reader may verify that

$$1 = 28 \cdot 17 + (-5) \cdot 95.$$

✣ If $\gcd(a, b) = 1$, as in Example 2.51(b), we say that a and b are **relatively prime**.

2.4.3 Euclidean Algorithm for Finding $\gcd(a, b)$

Suppose that $a > b > 0$ (otherwise interchange a and b). Then, by Theorem 2.43, we may write

$$a = k_1 b + r_1, \quad \text{where } k_1 \text{ and } r_1 \in \mathbb{Z}^+ \text{ and } 0 \le r_1 < b. \tag{2.11}$$

Now Theorem 2.45 tells us that if n divides a and b, then it must divide r_1, since $r_1 = a - k_1 b$. Similarly, if n divides b and r_1, then it must divide a. We see that the common divisors of a and b are the same as the common divisors of b and r_1, so

$$\gcd(a, b) = \gcd(b, r_1).$$

We now continue using Theorem 2.43 as follows:

$$\begin{array}{rll}
\text{divide } b \text{ by } r_1: & b = k_2 r_1 + r_2, & 0 \le r_2 < r_1 \\
\text{divide } r_1 \text{ by } r_2: & r_1 = k_3 r_2 + r_3, & 0 \le r_3 < r_2 \\
\text{divide } r_2 \text{ by } r_3: & r_2 = k_4 r_3 + r_4, & 0 \le r_4 < r_3 \\
 & \vdots & \vdots \\
\text{divide } r_{n-2} \text{ by } r_{n-1}: & r_{n-2} = k_n r_{n-1} + r_n, & 0 \le r_n < r_{n-1} \\
\text{divide } r_{n-1} \text{ by } r_n: & r_{n-1} = k_{n+1} r_n + r_{n+1}, & 0 \le r_{n+1} < r_n
\end{array} \tag{2.12}$$

Since $a > b > r_1 > r_2 > r_3 > r_4 > \cdots$, the remainder will eventually become zero, so at some point we have $r_{n+1} = 0$.
We now show that $r_n = \gcd(a, b)$. We saw previously that

$$\gcd(a, b) = \gcd(b, r_1).$$

Repeating this argument with b and r_1, we see that

$$\gcd(b, r_1) = \gcd(r_1, r_2).$$

Upon continuing, we have

$$\gcd(a, b) = \gcd(b, r_1) = \gcd(r_1, r_2) = \cdots = \gcd(r_{n-1}, r_n).$$

Since $r_{n-1} = k_{n+1}r_n$, we see that $\gcd(r_{n-1}, r_n) = r_n$. Hence $r_n = \gcd(a, b)$.

Example 2.52. Let a be 190 and b be 34. Then, using the Euclidean algorithm, we

$$\begin{aligned} \text{divide 190 by 34:} &\quad 190 = 5 \cdot 34 + 20 \\ \text{divide 34 by 20:} &\quad 34 = 1 \cdot 20 + 14 \\ \text{divide 20 by 14:} &\quad 20 = 1 \cdot 14 + 6 \\ \text{divide 14 by 6:} &\quad 14 = 2 \cdot 6 + 2 \\ \text{divide 6 by 2:} &\quad 6 = 3 \cdot 2 + 0. \end{aligned}$$

so that gcd(190, 34) is 2, the last of the divisors. □

Process of finding s and t: If $d = \gcd(a, b)$, we can find integers s and t such that $d = sa + tb$. The integers s and t can be found as follows. Solve the next-to-last equation in (2) for r_n:

$$r_n = r_{n-2} - k_n r_{n-1}. \tag{2.13}$$

Now solve the second-to-last equation in (2.12), $r_{n-3} = k_{n-1}r_{n-2} + r_{n-1}$, for r_{n-1}:

$$r_{n-1} = r_{n-3} - k_{n-1}r_{n-2}.$$

Substitute this expression in (2.13):

$$r_n = r_{n-2} - k_n[r_{n-3} - k_{n-1}r_{n-2}].$$

Continue to work up through the equations in (2.13) and (2.12), replacing r_i by an expression involving r_{i-1} and r_{i-2} and finally arriving at an expression involving only a and b.

Example 2.53. (a) Let $a = 190$ and $b = 34$ as in Example 2.52. Then

$$\begin{aligned} \gcd(190, 34) = 2 &= 14 - 2(6) \\ &= 14 - 2[20 - 1(14)] \quad (\text{since } 6 = 20 - 1 \cdot 14) \\ &= 3(14) - 2(20) \\ &= 3[34 - 1(20)] - 2(20) \quad (\text{since } 14 = 34 - 1 \cdot 20) \\ &= 3(34) - 5(190 - 5 \cdot 34) \quad (\text{since } 20 = 190 - 5 \cdot 34) \\ &= 28(34) - 5(190). \end{aligned}$$

Hence $s = -5$ and $t = 28$.
(b) Let $a = 108$ and $b = 60$. Then

$$\begin{aligned} \gcd(108, 60) = 12 &= 60 - 1(48) \\ &= 60 - 1[108 - 1(60)] \quad (\text{since } 48 = 108 - 1 \cdot 60) \\ &= 2(60) - 108. \end{aligned}$$

Hence $s = -1$ and $t = 2$.

Theorem 2.54. *If a and b are in $\mathbb{Z}^+$, then $gcd(a, b) = gcd(b, b \pm a)$.*

Proof. If c divides a and b, it divides $b \pm a$, by Theorem . Since $a = b - (b - a) = -b + (b + a)$, we see, also by Theorem 2.45, that a common divisor of b and $b \pm a$ also divides a and b. Since a and b have the same common divisors as b and $b \pm a$, they must have the same greatest common divisor. □

2.4.4 Least Common Multiple

✣ If a, b, and k are in $\mathbb{Z}^+$, and $a \mid k$, $b \mid k$, we say k is a **common multiple** of a and b.
✣ The smallest such k, call it c, is called the **least common multiple** or lcm, of a and b, and we write $c = \mathrm{lcm}(a, b)$.

Theorem 2.55. *If a and b are two positive integers, then $gcd(a, b) \cdot lcm(a, b) = ab$.*

Proof. Let $p_1, p_2, \ldots, p_k$ be all the prime factors of either a or b. Then we can write

$$a = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} \quad \text{and} \quad b = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$$

where some of the a_i and b_i may be zero. It then follows that

$$\gcd(a, b) = p_1^{\min(a_1, b_1)} p_2^{\min(a_2, b_2)} \cdots p_k^{\min(a_k, b_k)}$$

and

$$\mathrm{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \cdots p_k^{\max(a_k, b_k)}.$$

Hence

$$\begin{aligned} \gcd(a, b) \cdot \mathrm{lcm}(a, b) &= p_1^{a_1+b_1} p_2^{a_2+b_2} \cdots p_k^{a_k+b_k} \\ &= (p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}) \cdot (p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}) \\ &= ab. \end{aligned}$$

□

Example 2.56. Let $a = 540$ and $b = 504$. Factoring a and b into primes, we obtain

$$a = 540 = 2^2 \cdot 3^3 \cdot 5 \quad \text{and} \quad b = 504 = 2^3 \cdot 3^2 \cdot 7.$$

Thus all the prime numbers that are factors of either a or b are $p_1 = 2$, $p_2 = 3$, $p_3 = 5$, and $p_4 = 7$. Then $a = 2^2 \cdot 3^3 \cdot 5^1 \cdot 7^0$ and $b = 2^3 \cdot 3^2 \cdot 5^0 \cdot 7^1$. We then have

$$\begin{aligned} \gcd(540, 504) &= 2^{\min(2,3)} \cdot 3^{\min(3,2)} \cdot 5^{\min(1,0)} \cdot 7^{\min(0,1)} \\ &= 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^0 \\ &= 2^2 \cdot 3^2 \quad \text{or} \quad 36. \end{aligned}$$

Also,

$$\begin{aligned} \mathrm{lcm}(540, 504) &= 2^{\max(2,3)} \cdot 3^{\max(3,2)} \cdot 5^{\max(1,0)} \cdot 7^{\max(0,1)} \\ &= 2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 \quad \text{or} \quad 7560. \end{aligned}$$

Then $\gcd(540, 504) \cdot \mathrm{lcm}(540, 504) = 36 \cdot 7560 = 272{,}160 = 540 \cdot 504$.
Note: We can also compute $\gcd(540, 504)$ by the Euclidean algorithm and obtain the same result.
♦

Algorithm 2 Greatest Common Divisor (gcd), gcd(x, y)

```
while x ≠ y do
    if x > y then
        x ← x − y
    else
        y ← y − x
    end if
end while
return x
```

Algorithm to find greatest common divisor (gcd)

2.5 Modular Arithmetic and Its Applications

If $a \neq 0$ and b are nonnegative integers, Theorem 2.43 tells us that we can write $b = qa+r$, $0 \leq r < a$. Sometimes the remainder r is more important than the quotient q. Note that $0 \leq r < a$.

Example 2.57. If the time is now 4 o'clock, what time will it be 101 hours from now?
Solution: Let $a = 12$ and $b = 4 + 101$, or 105. Then we have $105 = 8 \cdot 12 + 9$. The remainder 9 answers the question. In 101 hours it will be 9 o'clock. ⧫

In this case we call a the **modulus** and write $b \equiv r \pmod{a}$, read "b is congruent to r mod a".

Example 2.58. (a) $29 \equiv 4 \pmod{5}$, since $29 = 5 \cdot 5 + 4$.

(b) $172 \equiv 7 \pmod{11}$, since $172 = 15 \cdot 11 + 7$.

(c) $3 \equiv 3 \pmod{6}$, since $3 = 0 \cdot 6 + 3$.

Note that if $b \equiv r \pmod{a}$, then $0 \leq r < a$, and $b - r$ is a multiple of a; that is, a divides $b - r$.
For each $n \in \mathbb{Z}^+$, we define a function f_n, the **mod-n** function, as follows: If z is a nonnegative integer, then $f_n(z) = r$, where $z \equiv r \pmod{n}$ and $0 \leq r < n$.

Example 2.59 (10). (a) $f_3(14) = 2$, because $14 = 4 \cdot 3 + 2$ and $14 \equiv 2 \pmod{3}$.

(b) $f_7(153) = 6$

2.5.1 Modular Arithmetic

Definition: Modular arithmetic involves integers and a modulus n, where numbers wrap around after reaching n.
We write:

$$a \equiv b \pmod{n}$$

if a and b leave the same remainder when divided by n.
Example:

$$17 \equiv 2 \pmod{5} \quad \text{since } 17 - 2 = 15 \text{ is divisible by } 5$$

Clock Analogy: 15 o'clock is equivalent to 3 o'clock on a 12-hour clock:

$$15 \equiv 3 \pmod{12}$$

Properties (mod n):

- **Addition:**

$$(a+b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$$

- **Multiplication:**

$$(a \cdot b) \bmod n = [(a \bmod n) \cdot (b \bmod n)] \bmod n$$

Example:

$$(23+19) \bmod 6 = 42 \bmod 6 = 0$$

Hence,

$$(23+19) \equiv 0 \pmod 6$$

2.6 Boolean Matrices:- Boolean matrix operation, Boolean Product of Two Matrices, Complement of Boolean Matrix

A Boolean matrix is an $m \times n$ matrix whose entries are either zero or one. We shall now define three operations on Boolean matrices that have some useful applications.
Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ Boolean matrices. We define $A \vee B = C = [c_{ij}]$, the **join** of A and B, by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both } 0. \end{cases}$$

and $A \wedge B = D = [d_{ij}]$, the **meet** of A and B, by

$$d_{ij} = \begin{cases} 1 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both } 1 \\ 0 & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0. \end{cases}$$

Note: These operations are only possible when A and B have the same size, just as in the case of matrix addition.

Example 2.60. Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

(a) Compute $A \vee B$.

(b) Compute $A \wedge B$.

Solution:

(a) Let $A \vee B = [c_{ij}]$. Then, since a_{43} and b_{43} are both 0, we see that $c_{43} = 0$. In all other cases, either a_{ij} or b_{ij} is 1, so c_{ij} is also 1. Thus

$$A \vee B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

(b) Let $A \wedge B = [d_{ij}]$. Then, since a_{11} and b_{11} are both 1, $d_{11} = 1$, and since a_{23} and b_{23} are both 1, $d_{23} = 1$. In all other cases, either a_{ij} or b_{ij} is 0, so $d_{ij} = 0$. Thus

$$A \wedge B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Finally, suppose that $A = [a_{ij}]$ is an $m \times p$ Boolean matrix and $B = [b_{ij}]$ is a $p \times n$ Boolean matrix. Notice that the condition on the sizes of A and B is exactly the condition needed to form the matrix product AB. We now define another kind of product.
The **Boolean product** of A and B, denoted $A \odot B$, is the $m \times n$ Boolean matrix $C = [c_{ij}]$ defined by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 \text{ and } b_{kj} = 1 \text{ for some } k, 1 \leq k \leq p \\ 0 & \text{otherwise.} \end{cases}$$

This multiplication is similar to ordinary matrix multiplication. The preceding formula states that for any i and j the element c_{ij} of $C = A \odot B$ can be computed in the following way (see Figure 1.20 and compare this with Figure 1.19).

1. Select row i of A and column j of B, and arrange them side by side.

2. Compare corresponding entries. If even a single pair of corresponding entries consists of two 1's, then $c_{ij} = 1$. If this is not the case, then $c_{ij} = 0$.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{p1} & b_{p2} & \cdots & b_{pj} & \cdots & b_{pn} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

where $c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{ip} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{pj} \end{bmatrix}$

Figure 1.20

If any corresponding pair of entries are both equal to 1, then $c_{ij} = 1$; otherwise $c_{ij} = 0$.
We can easily perform the indicated comparisons and checks for each position of the Boolean product. Thus, at least for human beings, the computation of elements in $A \odot B$ is considerably easier than the computation of elements in AB.

Example 2.61. Let $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$. Compute $A \odot B$.

Solution: Let $A \odot B = [e_{ij}]$. Then $e_{11} = 1$, since row 1 of A and column 1 of B each have a 1 as the first entry. Similarly, $e_{12} = 1$, since $a_{12} = 1$ and $b_{22} = 1$; that is, the first row of A and the second column of B have a 1 in the second position. In a similar way we see that $e_{13} = 1$. On the other hand, $e_{14} = 0$, since row 1 of A and column 4 of B do not have common 1's in any position. Proceeding in this way, we obtain

$$A \odot B = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

Theorem 2.62. *If A, B, and C are Boolean matrices of compatible sizes, then*

1(a) $A \vee B = B \vee A$.

(b) $A \wedge B = B \wedge A$.

2(a) $(A \vee B) \vee C = A \vee (B \vee C)$.

(b) $(A \wedge B) \wedge C = A \wedge (B \wedge C)$.

3(a) $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$.

(b) $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$.

3(a) $(A \odot B) \odot C = A \odot (B \odot C)$.

EXERCISE SET 1.5

1. Let $A = \begin{bmatrix} 3 & -2 & 5 \\ 4 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$, and $C = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 6 & -1 \\ 2 & 0 & 8 \end{bmatrix}$

   (a) What is a_{12}, a_{22}, a_{23}?

   (b) What is b_{11}, b_{31}?

   (c) What is c_{13}, c_{23}, c_{33}?

   (d) List the elements on the main diagonal of C.

2. Which of the following are diagonal matrices?

   (a) $A = \begin{bmatrix} 2 & 3 \\ 0 & 0 \end{bmatrix}$

   (b) $B = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

   (c) $C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

   (d) $D = \begin{bmatrix} 2 & 6 & -2 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

   (e) $E = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

3. If
$$\begin{bmatrix} a+b & c+d \\ c-d & a-b \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 10 & 2 \end{bmatrix}, \text{ find } a, b, c, \text{ and } d.$$

4. If
$$\begin{bmatrix} a+2b & 2a-b \\ 2c+d & c-2d \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ 4 & -3 \end{bmatrix}, \text{ find } a, b, c, \text{ and } d.$$

In Exercises 5 through 10, let

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 3 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 2 & 5 \\ 3 & 1 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} -3 & 2 \\ 4 & 1 \end{bmatrix},$$

$$E = \begin{bmatrix} 3 & 2 & -1 \\ 5 & 4 & -3 \\ 0 & 1 & 2 \end{bmatrix}, \text{ and } F = \begin{bmatrix} -2 & 3 \\ 4 & 5 \end{bmatrix}.$$

5. If possible, compute each of the following.

   (a) $C + E$

   (b) AB and BA

   (c) $CB + F$

   (d) $AB + DF$

6. If possible, compute each of the following.
   (a) $A(BD)$ and $(AB)D$
   (b) $A(C+E)$ and $AC+AE$
   (c) $FD+AB$
7. If possible, compute each of the following.
   (a) $EB+FA$
   (b) $A(B+D)$ and $AB+AD$
   (c) $(F+D)A$
   (d) $AC+DE$
8. If possible, compute each of the following.
   (a) A^T and $(A^T)^T$
   (b) $(C+E)^T$ and C^T+E^T
   (c) $(AB)^T$ and B^TA^T
   (d) $(B^TC)+A$
9. If possible, compute each of the following.
   (a) $A^T(D+F)$
   (b) $(BC)^T$ and C^TB^T
   (c) $(B^T+A)C$
   (d) $(D^T+E)F$
10. Compute D^3.
11. Let A be an $m \times n$ matrix. Show that $I_mA = AI_n = A$.
12. Let $A = \begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 \\ 3 & 4 \end{bmatrix}$. Show that $AB \neq BA$.
13. Let $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.
    (a) Compute A^3.
    (b) What is A^k?
14. Show $A0 = 0$ for any matrix A.
15. Show that $I_n^T = I_n$.
16. (a) Prove that if A has a row of zeros, then AB has a corresponding row of zeros.

(b) Prove that if B has a column of zeros, then AB has a corresponding column of zeros.

17. Show that the jth column of the matrix product AB is equal to the matrix product AB_j, where B_j is the jth column of B.

18. If 0 is the 2×2 zero matrix, find two 2×2 matrices A and B, with $A \neq 0$ and $B \neq 0$, such that $AB = 0$.

19. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, show that $A^2 = I_2$.

20. Determine all 2×2 matrices $A = \begin{bmatrix} 0 & a \\ b & c \end{bmatrix}$ such that $A^2 = I_2$.

21. Let A and B be symmetric matrices.

    (a) Show that $A + B$ is also symmetric.

    (b) Is AB also symmetric?

22. Let A be an $n \times n$ matrix.

    (a) Show that AA^T and $A^T A$ are symmetric.

    (b) Show that $A + A^T$ is symmetric.

23. Prove Theorem 3. [Hint: For part (c), show that the i, jth element of $(AB)^T$ equals the i, jth element of $B^T A^T$.]

24. In each part, compute $A \vee B$, $A \wedge B$, and $A \odot B$ for the given matrices A and B.

    (a) $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

    (b) $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

    (c) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

    (d) $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

    (e) $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

    (f) $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

25. (a) Show that $A \vee A = A$.

(b) Show that $A \wedge A = A$.

26. (a) Show that $A \vee B = B \vee A$.

    (b) Show that $A \wedge B = B \wedge A$.

27. (a) Show that $A \vee (B \vee C) = (A \vee B) \vee C$.

    (b) Show that $A \wedge (B \wedge C) = (A \wedge B) \wedge C$.

    (c) Show that $A \odot (B \odot C) = (A \odot B) \odot C$.

28. (a) Show that $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$.

    (b) Show that $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$.

29. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $n \times n$ matrices, and let $C = [c_{ij}]$ represent AB. Prove that if k is an integer and $k \mid a_{ij}$ for all i, j, then $k \mid c_{ij}$ for all i, j.

30. Let p be a prime number with $p > 2$, and let A and B be matrices all of whose entries are integers. Suppose that p divides all the entries of $A + B$ and all the entries of $A - B$. Prove that p divides all the entries of A and all the entries of B.